Cardia

An Intelligent Machine Learning-Based System for Heart Disease Prediction

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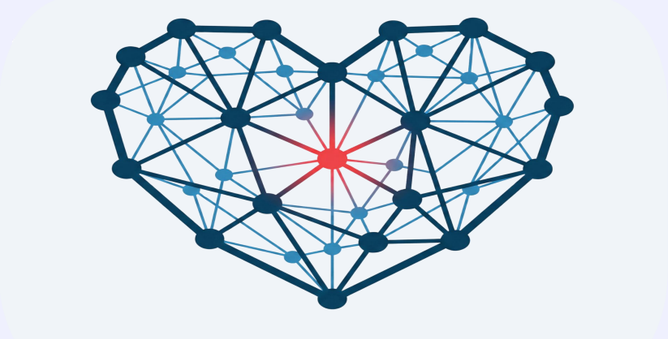

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Motivation

- The Global Challenge: Cardiovascular diseases remain the leading cause of mortality worldwide, where late detection is the primary barrier to survival.
- Early Intervention: Predictive systems empower healthcare providers to intervene preventively before irreversible heart damage occurs.
- Engineering the Solution: Cardia leverages Machine Learning to transform complex medical data into early-warning insights for risk assessment.
- The Vision: Shifting the healthcare paradigm from reactive treatment to proactive, data-driven prevention.

Research Problem

- Diagnostic Latency: Traditional methods often detect disease only after symptoms appear, limiting opportunities for early prevention.
- Data Complexity: Cardiovascular risk depends on high-dimensional, non-linear factors that are difficult to analyze consistently using conventional approaches.
- Resource Constraints: Comprehensive clinical screening is time-consuming and costly, highlighting the need for automated and scalable risk assessment tools.

Brief About The Idea

- Cardia is a full-stack intelligent platform for early cardiovascular risk assessment.
- Integrates a Machine Learning prediction engine with a modern web-based clinical interface.
- Enables healthcare providers to input patient data and receive real-time risk predictions.

Research Design

- Analytical Layer: Python & Scikit-learn for medical data pattern recognition.
- API Layer: RESTful services connecting ML engine with the user interface.
- Data Layer: MySQL database for secure clinical data storage.

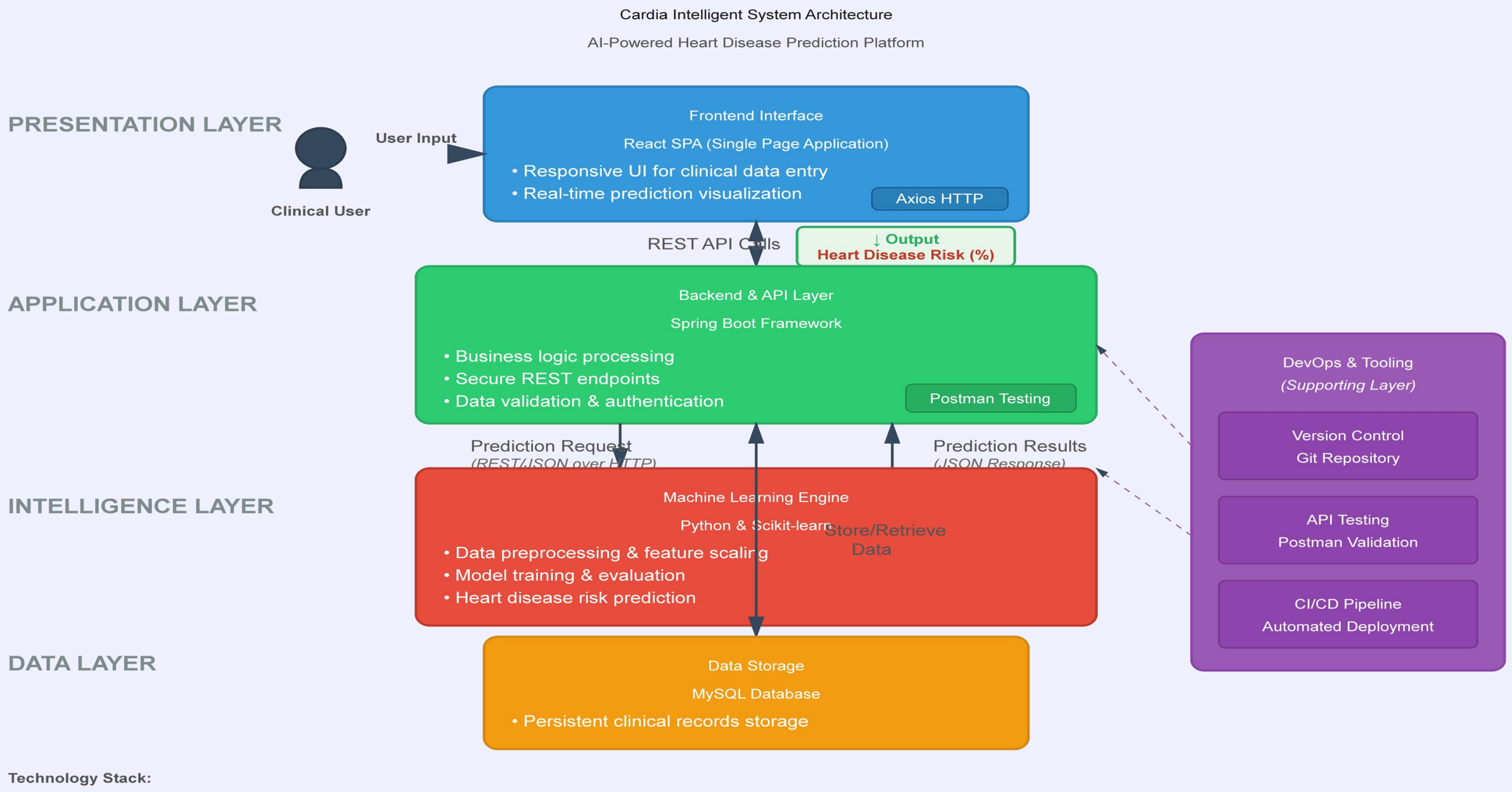
Implementation (Technical Pipeline)

Machine Learning Engine
- Built using Python & Scikit-learn for preprocessing, scaling, and model training.

Backend & API
- Developed with Spring Boot to manage business logic and secure REST endpoints.
- Postman used for API validation and testing.

Frontend Interface
- Responsive React SPA for clinical data entry and visualization.
- Axios enables real-time communication with backend services.

Data & DevOps
- MySQL for persistent clinical records.
- Git for version control and team collaboration.



Technology Stack:

Frontend: React.js, Axios
Backend: Spring Boot (Java)
ML: Python, Scikit-learn
Database: MySQL
Tools: Git, Postman